

Computer Organization and Architecture	
Lab #:	2
Lab Title:	Displays Even, Odd and average of Numbers
Submitted by:	
Names	Registration #
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Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Lab 02

Task No: 01

Write a program that gets a number from user and displays “Even” or “Odd. (If-else).

```
.data
odd_msg: .asciiz "Odd"
even_msg: .asciiz "Even"

main:
li $t0, 0      # initialize value
li $t1, 1      # initialize value

get_num:
li $v0, 5      # taking command from user
syscall
move $t2, $v0

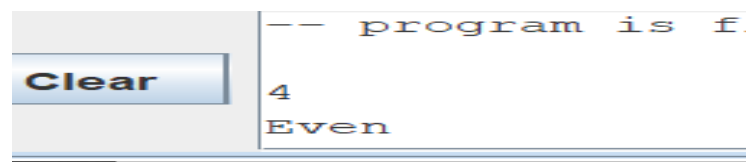
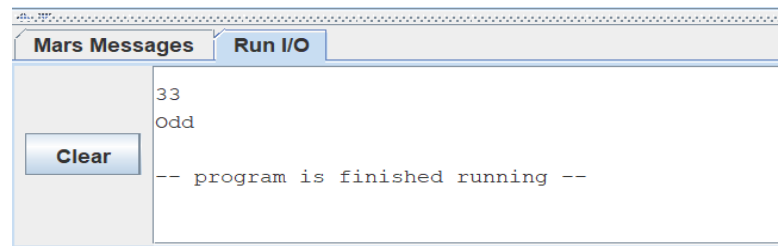
li $t3, 2      # divisor for even/odd check
rem $t4,$t2,$t3    # get remainder

beqz $t4,even  # if remainder is 0, jump to even
li $a0, 4      # print "Odd"
la $a0, odd_msg
j print_msg

even:
li $a0, 4      # print "Even"
la $a0, even_msg

print_msg:
li $v0, 4
syscall
li $v0, 10
syscall
```

Output:



Text Segment				
Bkpt	Address	Code	Basic	Source
☐	0x00400000	0x240200c	32-bit binary MIPS instruction	13: li \$v0, 5 # Load syscall code for reading integer
☐	0x00400004	0x0000000c	syscall	14: syscall
☐	0x00400008	0x00024021	addu \$8,\$0,\$2	15: move \$t0, \$v0 # Store user input in \$t0
☐	0x0040000c	0x31090001	andi \$9,\$8,0x00000001	18: andi \$t1, \$t0, 1 # Check the LSB to determine even or odd
☐	0x00400010	0x11200001	beq \$9,\$0,0x00000001	19: beq \$t1, \$zero, even # If LSB is 0, jump to even label
☐	0x00400014	0x0810000b	j 0x0040002c	20: j odd # If LSB is 1, jump to odd label
☐	0x00400018	0x24020004	addiu \$2,\$0,0x00000004	23: li \$v0, 4 # Load syscall code for printing string
☐	0x0040001c	0x3c011001	lui \$1,0x00001001	24: la \$a0, even msg # Load the address of the even message
☐	0x00400020	0x34240000	ori \$4,\$1,0x00000000	

Data Segment								
Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x6e657645	0x644f000a	0x00000a64	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

In Lab Task:

Take the marks of 5 students and displays its average by using concept of for loop in MIPS assembly

li \$t1,5

li \$t0 ,0

li \$t3 ,0 # initialize values

```

anylabel:
li $v0, 5    # taking command from user
syscall

move $t2, $v0
add $t3,$t3,$t2
add $t0,$t0,1
bne $t0,$t1,anylabel #jump back to anylabel
li $t4,5
div $t3,$t3,$t4
move $a0,$t3
li $v0,1
syscall
li $v0,10
syscall

```

Output:

